NanoGT Match Report: Choroideremia

Disease: Choroideremia (ORPHA:180)
Primary gene: CHM
Gene CDS: 1962 bp
Inheritance: X-linked recessive
Target tissues scored: retina

Interpretation

- At least one high-confidence precedent was found, but this is still a precedent match rather than a clinical-trial recommendation.
- Main review flags: AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone; Vector does not naturally cover all annotated disease tissues: retina; No direct tissue overlap; treat this as weak precedent unless route or modality is changed.

Disease Mechanism Evidence

Molecular mechanism: loss of function
Mechanistic detail: REP1 deficiency affecting retinal cells
Gene-addition compatibility: compatible
Preferred modality class: retinal gene addition
Evidence level/status: direct / source_linked_needs_review
Evidence summary: CHM loss of function is a retinal gene-addition target if viable retina remains
Evidence source: OMIM CHM gene entry

Study-Level Limitations

- Catalog-relative ranking: current catalog contains 21 precedent programs and 8 vectors, so absence of a strong match is not proof that no therapy is possible.
- Modality coverage is limited mainly to AAV and lentiviral precedents; dual-AAV, LNP/mRNA, genome editing, ASO, and transplant-enabling strategies are not fully represented.
- Few catalog vectors cover the annotated disease tissue(s): AAV2.
- Endpoint readiness: retinal diseases often have measurable endpoints such as OCT, ERG, visual acuity, visual fields, or mobility testing, but genotype-specific progression still needs confirmation.

Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	Luxturna	AAV2	8.0/10	High	approved
2	Strimvelis	LV	7.4/10	Medium	approved
3	CPCB-RPE1	AAV8	7.3/10	Medium	phase2/3
4	Skysona	LV	7.0/10	Medium	approved
5	BMN 307	AAV5	7.0/10	Medium	phase2

Match #1: Luxturna

Precedent disease: Leber congenital amaurosis type 2
Vector: AAV2
Tissue target: retina/RPE
Composite score: 8.0 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	0.50	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	1.00	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	0.80	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**7.95**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 1962bp / cargo 4700bp (42% utilized)
- Vector tropism plus precedent target match: retina
- Both intracellular proteins
- LOF inheritance — compatible for gene replacement
- Pathway match: retinal_visual_cycle
- Disease mechanism: loss of function — REP1 deficiency affecting retinal cells
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: CHM loss of function is a retinal gene-addition target if viable retina remains
- Mechanism source: OMIM CHM gene entry (https://omim.org/entry/300390)
- Approval status: approved
- Vector immunogenicity (AAV2): very high (~55%) — majority of patients may be ineligible; major trial design challenge

- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF-β2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010
- ORGANELLE TARGETING: COMPATIBLE — CHM standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #2: Strimvelis

Precedent disease: ADA-SCID
Vector: LV
Tissue target: hematopoietic
Composite score: 7.4 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	1.00	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	0.80	1.0	Established delivery route to target tissue

Dimension	Score	Max	What it measures
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**7.38**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 1962bp / cargo 8000bp (25% utilized)
- No tissue overlap (disease: ['retina'], vector: ['hematopoietic'])
- Both intracellular proteins
- LOF inheritance — compatible for gene replacement
- Different pathway (retinal_phototransduction vs immune_hematopoietic)
- Disease mechanism: loss of function — REP1 deficiency affecting retinal cells
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: CHM loss of function is a retinal gene-addition target if viable retina remains
- Mechanism source: OMIM CHM gene entry (https://omim.org/entry/300390)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF-β2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010
- ORGANELLE TARGETING: COMPATIBLE — CHM standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Vector does not naturally cover all annotated disease tissues: retina
- No direct tissue overlap; treat this as weak precedent unless route or modality is changed

Match #3: CPCB-RPE1

Precedent disease: Achromatopsia
Vector: AAV8
Tissue target: retina/photoreceptor
Composite score: 7.3 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.50	2.0	Vector naturally reaches disease target tissue

Dimension	Score	Max	What it measures
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.70	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	1.00	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	0.80	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**7.33**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 1962bp / cargo 4700bp (42% utilized)
- Precedent target match: retina
- Protein class mismatch
- LOF inheritance — compatible for gene replacement
- Pathway match: retinal_phototransduction
- Disease mechanism: loss of function — REP1 deficiency affecting retinal cells
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: CHM loss of function is a retinal gene-addition target if viable retina remains
- Mechanism source: OMIM CHM gene entry (https://omim.org/entry/300390)
- Approval status: phase2/3
- Vector immunogenicity (AAV8): high (~30%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF-β2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010
- ORGANELLE TARGETING: COMPATIBLE — CHM standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no

additional organelle-import steps required.

Manual Review Flags

- Vector does not naturally cover all annotated disease tissues: retina
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #4: Skysona

Precedent disease: Cerebral adrenoleukodystrophy
Vector: LV
Tissue target: hematopoietic/CNS
Composite score: 7.0 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	1.00	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	0.80	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**7.05**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 1962bp / cargo 8000bp (25% utilized)
- No tissue overlap (disease: [‘retina’], vector: [‘hematopoietic’])
- Protein class mismatch
- Inheritance match (X-linked recessive <-> XL)
- Different pathway (retinal_phototransduction vs peroxisomal)

- Disease mechanism: loss of function — REP1 deficiency affecting retinal cells
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: CHM loss of function is a retinal gene-addition target if viable retina remains
- Mechanism source: OMIM CHM gene entry (https://omim.org/entry/300390)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF-β2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010
- ORGANELLE TARGETING: COMPATIBLE — CHM standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Vector does not naturally cover all annotated disease tissues: retina
- No direct tissue overlap; treat this as weak precedent unless route or modality is changed

Match #5: BMN 307

Precedent disease: Phenylketonuria
Vector: AAV5
Tissue target: liver
Composite score: 7.0 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.60	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?

Dimension	Score	Max	What it measures
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	1.00	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	0.80	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**6.95**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 1962bp / cargo 4700bp (42% utilized)
- No tissue overlap (disease: ['retina'], vector: ['liver', 'lung', 'CNS'])
- Both intracellular proteins
- LOF inheritance — compatible for gene replacement
- Different pathway (retinal_phototransduction vs amino_acid_metabolism)
- Disease mechanism: loss of function — REP1 deficiency affecting retinal cells
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: CHM loss of function is a retinal gene-addition target if viable retina remains
- Mechanism source: OMIM CHM gene entry (https://omim.org/entry/300390)
- Approval status: phase2
- Vector immunogenicity (AAV5): low (~9%) — most patients eligible; minimal screening burden
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: highest immune privilege — blood-retinal barrier + FasL + TGF-β2; minimal T-cell clearance risk
- Promoter availability: VMD2, RPGR, GRK1, CRX, IRBP — multiple validated retinal promoters used in Luxturna, GS010, CPCB-RPE1
- Route of administration: Subretinal/intravitreal injection — specialist ophthalmic procedure; established in Luxturna and GS010
- ORGANELLE TARGETING: COMPATIBLE — CHM standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Vector does not naturally cover all annotated disease tissues: retina
- No direct tissue overlap; treat this as weak precedent unless route or modality is changed
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone